

Sulaeman Aloradi

Am Jesuitenhof 3, 53117 Bonn, Germany

Email: snedal99@gmail.com | Phone: +49 1575 4221331

LinkedIn: [linkedin.com/in/sulaeman-aloradi-3927a5212](https://www.linkedin.com/in/sulaeman-aloradi-3927a5212)

GitHub: github.com/Sulaiman-Nedal

Portfolio: sulaiman-nedal.github.io/Portfolio



Profile

M.Sc. Computer Science candidate at the University of Bonn, about to graduate, with hands-on research across DLR, Fraunhofer ISE, UKB, and b-it. Specialising in generative AI, transformer fine-tuning, and end-to-end NLP pipelines. Seeking full-time roles in Data Science, NLP, or Machine Learning in Germany.

Work Experience

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| 07/2026 - Present | Research Assistant (Wissenschaftliche Hilfskraft), Computational Biology
<i>Universitätsklinikum Bonn (UKB), Bonn</i> <ul style="list-style-type: none">• Protein structure prediction using AlphaFold2 for biomedical research at University Hospital Bonn.• Analyse predicted 3D structures to support research on disease-associated protein variants and drug-target interactions.• Collaborate with clinical researchers to integrate computational structure predictions into experimental biological workflows. |
| 04/2026 - Present | Working Student, Master's Thesis, NLP Research
<i>Deutsches Zentrum für Luft- und Raumfahrt (DLR), Jena</i> <ul style="list-style-type: none">• Developing an unsupervised NLP pipeline for fine-grained topic classification of a multilingual web corpus exceeding 685,000 documents, as part of the OpenSearch@DLR 2.0 initiative.• Applying multilingual embedding models (Sentence-BERT family), BERTopic, LDA, and lbl2vec to cluster and semantically label the corpus across multiple languages.• Evaluating topic and cluster quality using silhouette scores, coherence metrics, and structured human assessment protocols.• Collaborating within an interdisciplinary team spanning DLR Jena and the University of Bonn. |
| 11/2025 - 06/2026 | Working Student, Scientific Research
<i>Fraunhofer Institute for Solar Energy Systems (ISE), Freiburg, Battery Cell Technology Group</i> <ul style="list-style-type: none">• Maintained and further developed evaluation and processing software for battery measurement data within the Battery Cell Technology group.• Developed Python data pipelines and LLM integrations.• Managed version control of the developed software using GitLab, including branching and merge-request workflows.• Authored technical documentation for the software development, ensuring reproducibility and maintainability for the research team. |
| 04/2025 - 09/2025 | Working Student, NLP
<i>Bonn-Aachen International Center for Information Technology (b-it), Bonn</i> <ul style="list-style-type: none">• Fine-tuned transformer models (GPT-2, SciFive, BioBERT) for named entity recognition, question answering, natural language inference, and relation extraction on biomedical text.• Built and benchmarked QA and information-extraction pipelines that outperformed BERT and BioBERT baselines across multiple evaluation datasets. |

- Designed and evaluated retrieval-augmented generation (RAG) systems for biomedical knowledge retrieval, comparing dense vs. sparse retrieval strategies.
- Ran training and evaluation workloads on HPC/SLURM clusters; evaluated LLMs for accuracy, inference latency, and adversarial robustness.

Education

04/2023 - Present	Master of Science (M.Sc.) in Computer Science <i>University of Bonn (Rheinische Friedrich-Wilhelms-Universität Bonn), Bonn, Germany</i> Thesis: Unsupervised topic modelling for multilingual web corpora. Focus: NLP, Machine Learning.
09/2018 - 06/2022	Bachelor of Science (B.Sc.) in Computer Science <i>King Abdulaziz University, Jeddah, Saudi Arabia</i> Thesis: Recommendation system for educational material quality assessment.

Projects

Maze Navigation with MLM-U Multi-Step Prediction (b-it)

Implemented a transformer trained with a multi-step look-ahead (MLM-U) objective for maze navigation, based on the Facebook Research paper "Transformers Can Navigate Mazes With Multi-Step Prediction"; achieved higher accuracy and faster convergence than standard next-token prediction.

LLM-Based Communication Agent for Human-Robot Interaction (University of Bonn)

Built an LLM-powered speech function that lets a robot hold goal-directed conversations: generating responses, validating user input, and adapting until the goal is reached.

Churn Prediction from Mobile Game Telemetry (University of Bonn)

Built an end-to-end ML pipeline engineering gameplay features from two mobile games and comparing logistic regression, decision trees, and random forests to predict churn within a 10-day horizon, including feature-importance analysis.

Skills

Programming	Python, Java, SQL
ML / NLP	PyTorch, Hugging Face, LLM Fine-tuning, scikit-learn, NLTK, RAG, BERTopic
Bioinformatics	AlphaFold, protein structure prediction
Data	NumPy, Pandas, Matplotlib, Seaborn
Infrastructure	Linux, HPC/SLURM, Git, ROS2

Languages

Arabic	Native
English	C1
German	B2

Hobbies

Jogging, Board Games, Creative Writing